numbers between iterations to give a small computational saving. We can see that only one vector of random numbers is generated per iteration, with the new vector of random numbers being used in the following iteration. This gives some correlation between steps that complicates our analysis a little.

If we wish to find the perfect sampling bias in the case of the one-dimensional harmonic oscillator, we can apply this technique to find the invariant averages under this scheme. We have

$$\Psi^{\mathrm{BBK}^*} = \begin{bmatrix} 1-\frac{h^2\Omega^2}{2m} & \frac{h}{m}\left(1-\frac{\gamma h}{2}\right) \\ \frac{h\Omega^2}{2m}\left(\frac{h^2\Omega^2-4m}{2+\gamma h}\right) & \frac{(2-\gamma h)\left(2m-h^2\Omega^2\right)}{2m(2+\gamma h)} \end{bmatrix},$$

$$\mu_n^{\mathrm{BBK}^*} = \begin{bmatrix} \sqrt{\frac{k_\mathrm{B}\mathrm{T}\gamma h^3}{2m}}\mathrm{R}_n \\ \frac{\sqrt{2k_\mathrm{B}\mathrm{T}\gamma hm}}{2+\gamma h}\left(\left(1-\frac{h^2\Omega^2}{2m}\right)\mathrm{R}_n+\mathrm{R}_{n+1}\right) \end{bmatrix} = \begin{bmatrix} \kappa_1\mathrm{R}_n \\ \kappa_2\mathrm{R}_n+\kappa_3\mathrm{R}_{n+1} \end{bmatrix},$$

using the notation in (7.7). Taking expectations of the resulting equations, we can compute the values of the constants in (7.8)–(7.10) and solve to find the long-time averages. Care must be taken however, as unlike the schemes using the ABO splitting we must include some correlation between steps. Denoting the entries of Ψ^{BBK^*} by ψ_{ij} we have

$$\begin{aligned}\mathbb{E}(\mu_{n,2}p_n) &= \mathbb{E}\left(\left(\kappa_2\mathrm{R}_n+\kappa_3\mathrm{R}_{n+1}\right)\left(\psi_{21}q_{n-1}+\psi_{22}p_{n-1}+\kappa_2\mathrm{R}_{n-1}+\kappa_3\mathrm{R}_n\right)\right), \\ &= \kappa_2\kappa_3.\end{aligned}$$

This additional quantity will slightly change the expectation of (7.9) compared to the standard BBK method, and hence the overall averages of the scheme. Additionally it is easily shown that

$$\mathbb{E}(\mu_{n,1}q_n) = \mathbb{E}(\mu_{n,2}q_n) = 0, \quad \mathbb{E}(\mu_{n,1}p_n) = \kappa_1\kappa_3,$$

with

$$\mathbb{E}(\mu_{n,1}^2) = \kappa_1^2, \quad \mathbb{E}(\mu_{n,1}\mu_{n,2}) = \kappa_1\kappa_2, \quad \mathbb{E}(\mu_{n,2}^2) = \kappa_2^2+\kappa_3^2.$$

Thus taking expectations of (7.8)–(7.10) and substituting in the constants gives

$$\begin{aligned}\langle q^2\rangle_h^{\mathrm{BBK}^*} &= \psi_{11}^2\langle q^2\rangle_h^{\mathrm{BBK}^*}+\psi_{12}^2\langle p^2\rangle_h^{\mathrm{BBK}^*}+2\psi_{11}\psi_{12}\langle qp\rangle_h^{\mathrm{BBK}^*}+2\psi_{12}\kappa_1 2\psi_{12}\kappa_1\kappa_3 \\ &\quad +\kappa_1^2, \\ \langle p^2\rangle_h^{\mathrm{BBK}^*} &= \psi_{21}^2\langle q^2\rangle_h^{\mathrm{BBK}^*}+\psi_{22}^2\langle p^2\rangle_h^{\mathrm{BBK}^*}+2\psi_{21}\psi_{22}\langle qp\rangle_h^{\mathrm{BBK}^*}+2\psi_{22}\kappa_2 2\psi_{22}\kappa_2\kappa_3 \\ &\quad +\kappa_2^2+\kappa_3^2,\end{aligned}$$

$$\langle qp\rangle_h^{\mathrm{BBK}^*} = \psi_{11}\psi_{21}\langle q^2\rangle_h^{\mathrm{BBK}^*} + \psi_{12}\psi_{22}\langle p^2\rangle_h^{\mathrm{BBK}^*} + (\psi_{11}\psi_{22} + \psi_{12}\psi_{21})\langle qp\rangle_h^{\mathrm{BBK}^*} + \kappa_1\kappa_2 + \psi_{12}\kappa_2\kappa_3 + \psi_{22}\kappa_1\kappa_3.$$

Finally, solving the simultaneous equations gives the stationary observed averages as

$$\langle q^2\rangle_h^{\mathrm{BBK}^*} = \frac{k_\mathrm{B}\mathrm{T}}{\Omega^2}\left(1 - \frac{\Omega^2 h^2}{4m}\right)^{-1}, \quad \langle p^2\rangle_h^{\mathrm{BBK}^*} = k_\mathrm{B}\mathrm{T}m\left(1 + \frac{\gamma h}{2}\right)^{-1},$$
$$\langle qp\rangle_h^{\mathrm{BBK}^*} = 0.$$

For more general methods requiring multiple correlated random numbers, a further history than just one step is required to evaluate the expectation correctly.

7.5 Errors in Configurational Quantities for the Perturbed Harmonic Oscillator

We can consider applying the analysis of Sect. 7.4 to a more complicated example, however in general for nonlinear systems we cannot adopt the same strategy by taking expectations of the update equations, as we do not end up with a closed system of linear equations to solve (the force term now involves terms of higher orders). However, if the nonlinear terms in the force are preceded by a small constant, we can truncate the equations to provide an approximation to the invariant behavior. We consider sampling canonically the Hamiltonian

$$H(q,p) = p^2/2 + q^2/2 + \varepsilon q^4/4, \tag{7.13}$$

using Langevin dynamics, where $q, p \in \mathbb{R}$, with a small nonlinearity parameter $\varepsilon > 0$. As the canonical momentum distribution is always Gaussian (and hence trivial to sample), we will be far more concerned with the behavior of the (more interesting) distribution of position. The system can be thought of as the harmonic oscillator with a small nonlinear perturbation term, so the canonical configurational distribution will remain "close" to a Gaussian in some sense (this potential energy function is equivalent to an unforced Duffing Oscillator with no damping). In the limit as $\varepsilon \to 0$, we would expect the averages to converge to the results of Table 7.1.

We consider estimating the value of $\langle q^2\rangle_h$ by truncating the equations governing the evolution of the averages in powers of ε (to ensure closure of the resulting simultaneous equations). We examine the expected behavior of the average in the case of the four second-order schemes listed in Table 7.1, although we may conduct this analysis for any similar scheme.